Auto-Restart / Auto-Stop — Sequence of Operations

Overview

The Auto-Restart system automatically stops and restarts a gas compressor based on suction/discharge pressure conditions, with configurable retry logic and an Auto/Local mode system for personnel safety lockout.

Configuration Parameters (Global Settings Page 11)

Parameter	Key	Description	Default
Enable Auto Restart	`AutoRestartEnabled`	Master enable for entire AR system	`0` (off)
Restart on Any Fault	`ARRestartAnyFault`	Auto-restart after any fault, not just pressure	`0` (off)
Max Retries	`AutoRestartMaxRetries`	Max restart attempts before permanent fault	`3`
Retry Delay (mins)	`AutoRestartRetryDelay`	Standby countdown before each restart attempt	`60 mins`
Reset Time (mins)	`AutoRestartResetTime`	Run time required to reset retry counter to 0	`20 mins`
Low Stop Input	`ARSuctTerm` / `ARSuctChan`	Suction pressure input channel	None
Stop/Standby Below	`ARSuctLoStandby`	Suction pressure below this → stop & standby	—
Restart Above	`ARSuctHiStandby`	Suction must rise above this to restart	—
High Stop Input	`ARDischTerm` / `ARDischChan`	Discharge pressure input channel	None
Stop/Standby Above	`ARDischHiStandby`	Discharge pressure above this → stop & standby	—
Restart Below	`ARDischLoStandby`	Discharge must drop below this to restart	—
Switch Input	`ARAutoLocalTerm` / `ARAutoLocalChan`	Physical auto/local selector switch	None
Default System Mode	`ARPowerUpMode`	Mode on power up: `"auto"` or `"local"`	`"local"`

Auto/Local Mode System

Purpose

Provides a personnel safety lockout mechanism. When in **Local** mode, the unit will not auto-start. Technicians set the unit to Local before working on it.

Mode Determination

The `AutoLocal` virtual variable holds the current mode (`"Auto"` or `"Local"`). It is determined by:

1. Power-Up Default

On first boot after power cycle, `AutoLocal` is set based on `ARPowerUpMode` :

- `"auto"` → unit starts in Auto mode
- `"local"` (default) → unit starts in Local mode

2. Physical Switch Input (if configured)

- **Closed (value > 0.5)** → Auto
- **Open (value ≤ 0.5)** → Local (fail-safe: broken wire = Local)
- Switch state change overrides current mode
- Switch in Local position **always forces Local**

3. Manual Stop Override

- If a **Manual Stop** (fault type 4) or **Remote Stop** (fault type 14) occurs while in Auto mode, the system automatically reverts to **Local** mode
- This prevents the unit from auto-restarting after an intentional stop
- The operator must explicitly switch back to Auto to re-enable auto-restart

Mode Evaluation Order (each tick)

```
1. If physical switch configured:
   a. If switch state changed → override AutoLocal to match
   b. If switch is open (Local) → force AutoLocal = "Local"
2. If faulted AND in Auto AND fault type is 4 or 14 → force Local
```

Standby Entry (Auto-Stop)

Trigger Conditions

The system checks pressure values every tick (via `checkAutoRestart()`). Standby is triggered when **all** of the following are true:

1. `AutoRestartEnabled == 1`
2. `AutoLocal == "Auto"`
3. Unit is running (state 1–8) or ready (state 0, not faulted)
4. Retry count < Max Retries

Low Pressure Stop (Suction)

- Suction channel configured (`ARSuctTerm` / `ARSuctChan` > 0)
- `ARSuctLoStandby` is set
- Current suction value **drops below** `ARSuctLoStandby`
- Cause: `"lowSuction"`

High Pressure Stop (Discharge)

- Discharge channel configured (`ARDischTerm` / `ARDischChan` > 0)

- `ARDischHiStandby` is set
- Current discharge value **rises above** `ARDischHiStandby`
- Cause: `"highDischarge"`

Auto Mode Entry (State 0)

- Unit is at state 0 (ready), not faulted, no start pending
- Both Auto mode conditions met
- This handles the case where both are in Auto and the unit is simply stopped
- Cause: `"autoMode"`

Standby Entry Sequence

```
1. Stop the unit (call stop())
2. Set state to 10 (Standby)
3. Record standby metadata:
   - standbyType: "Low" or "High" or "Auto"
   - standbyLabel: channel label + terminal:channel
   - standbyTerm / standbyCh: source channel
   - arStandbyCause: "lowSuction", "highDischarge", or "autoMode"
4. Set arFaultRetryPending = true
5. Reset debounce state
6. Log standby event (event type 8, pause-circle amber icon)
   - Includes channel value and setpoint
7. Start standby countdown timer (AutoRestartRetryDelay minutes)
```

Max Retries Exceeded

If retry count ≥ Max Retries when entering standby:

```
1. Set permanent fault (type 10)
   - Low: "Low <label> ARS fail SP:<setpoint>"
   - High: "High <label> ARS fail SP:<setpoint>"
   - Auto: "<type> <label> ARS fail"
2. Stop unit, set state to 0
3. Unit stays faulted until manual reset
4. Retry counter remains at max (only resets after running for ResetTime minutes)
```

Standby Countdown (State 10)

During Countdown

- Status string shows: `"Standby <type> <label> <Xm Xs> [Auto]"` or `"[Local]"`
- Timer counts down from `AutoRestartRetryDelay` minutes

Mode Change During Countdown

- If mode switches to **Local** during countdown:
  - Status shows `"Standby – Local mode"`
  - Countdown continues but unit will **not** restart
  - Cancels auto-restart with custom fault `"– Auto-Restart Cancelled"`

Timer Expiry — Pre-Start Validation

When the standby timer reaches 0:

```
1. Check if still in Auto mode
   - If Local → cancel, return

2. Validate restart conditions based on cause:
   a. lowSuction:
      - Read current suction value
      - Must be ABOVE ARSuctHiStandby (Restart Above threshold)
      - If not met → show "Waiting: pressure X.X <= Y", keep waiting
   b. highDischarge:
      - Read current discharge value
      - Must be BELOW ARDischLoStandby (Restart Below threshold)
      - If not met → show "Waiting: pressure X.X >= Y", keep waiting
   c. autoMode:
      - Always passes (no pressure check needed during first start)

3. If conditions NOT met:
   - Reset debounce, keep waiting
   - Does NOT count as a retry attempt
   - Keeps checking each tick until conditions are met

4. If conditions ARE met:
   - Start 5-second debounce timer
   - Status: "Conditions met, restarting in Xs"
   - After 5 seconds of sustained conditions:
     → Proceed to restart
```

Restart Attempt

When conditions are met and debounce completes:

```
1. Set state to 0 (Ready)
2. Set autoReset = true, resetFlag = true (clears any fault)
3. Set startPressed = true (triggers start sequence)
4. Log auto-restart attempt (event type 9, play-circle cyan icon)
   - "Auto-Restart attempt N"
5. Increment retry counter (asRetries + 1)
6. Unit proceeds through normal start sequence (states 1 → 2 → ... → 8)
```

Fault During Restart Attempt

If the unit faults during the start sequence (e.g., pressure drops again mid-crank):

With "Restart on Any Fault" ENABLED (default)

```
1. Unit faults → goes to state 0 with fault flag
2. checkAutoRestart() sees: state 0, faulted, Auto mode, ARRestartAnyFault=1
3. Check fault type:
   - Fault type 4 (Manual Stop) or 14 (Remote Stop) → SKIP (intentional)
```

```
   - Any other fault type → proceed
4. Check retry count < Max Retries
   - If yes: clear fault (resetFlag + autoReset), log event, re-enter standby cycle
   - If no: stop retrying, unit stays faulted
```

With "Restart on Any Fault" DISABLED

```
1. Unit faults → goes to state 0 with fault flag
2. checkAutoRestart() sees the fault but ARRestartAnyFault=0
3. Sets retry count to max (effectively stops retrying)
4. Unit stays faulted until manual reset
```

Retry Counter Reset

If the unit successfully starts and runs for `AutoRestartResetTime` minutes:

```
1. checkAutoRestart() checks: asRetries > 0 AND elapsedTime > ResetTime AND state > 0
2. Resets asRetries to 0
3. Full retry budget is restored for next standby event
```

Event Log Entries

Event Type	Icon	Color	Description
8 (Standby)	pause-circle	Amber (#e8b339)	Entering standby state, fault retry during AR cycle
9 (Auto Restart)	play-circle	Cyan (#13c2c2)	Restart attempt initiated

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Dashboard Display

Status Strings (State 10)

Scenario	Status String
Countdown active, Auto mode	`"Standby Low suction 4m32s [Auto]"`
Timer expired, conditions not met	`"Waiting: pressure 25.0 <= 30"`
Timer expired, conditions met	`"Conditions met, restarting in 4s"`

Virtual Variables

Variable	Purpose
`AutoLocal`	Current mode: "Auto" or "Local"
`asr` / `_asRetries`	Current retry count (for dashboard display)

`_standbyLoHi`	0 = low, 1 = high (standby type indicator)
`_standbyTerm` / `_standbyCh`	Standby source channel

Complete Lifecycle Example

Scenario: Low suction pressure causes standby, then recovers

```
Time 00:00 — Unit running (state 8), Auto mode, suction = 50 PSI
             Stop Below = 40, Restart Above = 45, Retry Delay = 5 min

Time 01:00 — Suction drops to 38 PSI (below 40)
             → checkAutoRestart() detects: val(38) < ARSuctLoStandby(40)
             → enterAutoRestartStandby("lowSuction")
             → Unit stops, enters state 10
             → Log: "Standby: Low suction T1:IN5 value: 38" (type 8)
             → Countdown starts: 5 minutes

Time 01:00–06:00 — Standby countdown
             → Status: "Standby Low suction T1:IN5 4m30s [Auto]"
             → Suction still at 38 PSI

Time 06:00 — Timer expires, check restart conditions
             → Suction = 38 PSI, threshold = 45
             → 38 < 45 → conditions NOT met
             → Status: "Waiting: pressure 38.0 <= 45"
             → No retry consumed, keeps checking

Time 08:00 — Suction recovers to 46 PSI
             → 46 > 45 → conditions met!
             → Start 5-second debounce
             → Status: "Conditions met, restarting in 5s"

Time 08:05 — Debounce complete, suction still at 46
             → Set state 0, clear fault, press start
             → Log: "Auto-Restart attempt 1" (type 9)
             → asRetries = 1
             → Unit begins start sequence (states 1 → 2 → ... → 8)

Time 08:30 — Unit running at state 8
             → If runs for 20 min (Reset Time) → asRetries resets to 0
```

Scenario: Fault during restart, retry through fault

```
Time 00:00 — Unit in standby (state 10), countdown expired, conditions met

Time 00:05 — Debounce complete, restart attempt 1 initiated
             → Unit starts, enters state 3 (cranking)

Time 00:15 — Fault occurs during cranking (e.g., overcrank timeout, type 5)
```

```
             → Unit faults, goes to state 0
             → ARRestartAnyFault = 1 (enabled)
             → Fault type 5 ≠ 4 and ≠ 14 → eligible for retry
             → asRetries(1) < maxRetries(3) → clear fault
             → Log: "Auto-Restart: restarting after fault (type 5)" (type 8)
             → Unit returns to ready state, re-enters standby on next tick

Time 00:16 — checkAutoRestart() sees state 0, Auto, not faulted, retries < max
             → enterAutoRestartStandby("autoMode")
             → New standby countdown begins

Time 05:16 — Timer expires, conditions met (autoMode = always passes)
             → Restart attempt 2 initiated

             ... (repeats until success or max retries)

Time XX:XX — asRetries reaches 3 (max)
             → enterAutoRestartStandby sees retries >= max
             → Permanent fault: "Auto ARS fail"
             → Unit stays faulted, requires manual reset
```

Scenario: Manual stop during auto-restart cycle

```
Time 00:00 — Unit in standby (state 10), countdown active, Auto mode

Time 02:00 — Operator presses STOP button
             → sendStop() fires (state 10 included in stop conditions)
             → buttons() processes stop: fault type 4 (Manual Stop)
             → checkAutoLocal() sees fault type 4 while in Auto
             → Forces AutoLocal = "Local"
             → Unit will NOT auto-restart (Local mode)
             → Operator must switch back to Auto to re-enable
```